

REAL-TIME AIR MONITORING SYSTEM

AGENDA

Abstract

Project Outcome

Project Impact

Block Diagram

Brief Description of each block

Flow Chart of Air Monitoring System

Hardware and Software used in this Project

Experimental Setup

Result

Future Scope of Work

ABSTRACT

Considering the significance of air quality on human lives, the World Health Organization (WHO) has developed guidelines for reducing the health effects of air pollution on public health by setting the limits of the concentrations of various air pollutants.

In view of the ever-increasing pollution sources with toxic chemicals, our system have the facilities to detect and quantify these sources rapidly. The system uses low cost air-quality monitoring nodes comprises of low cost semiconductor gas sensors. This system measures concentrations of gases such as CO, CO₂, NH₃ and other harmful gases collectively using semiconductor sensors. The sensors will gather the data of various environmental parameters and provide it to raspberry pi which act as a base station. For real time monitoring of these data Raspberry Pi is connected to ThingSpeak cloud platform.

PROJECT OUTCOME

Prototype of Air Monitoring System

The prototype will comprises the following:

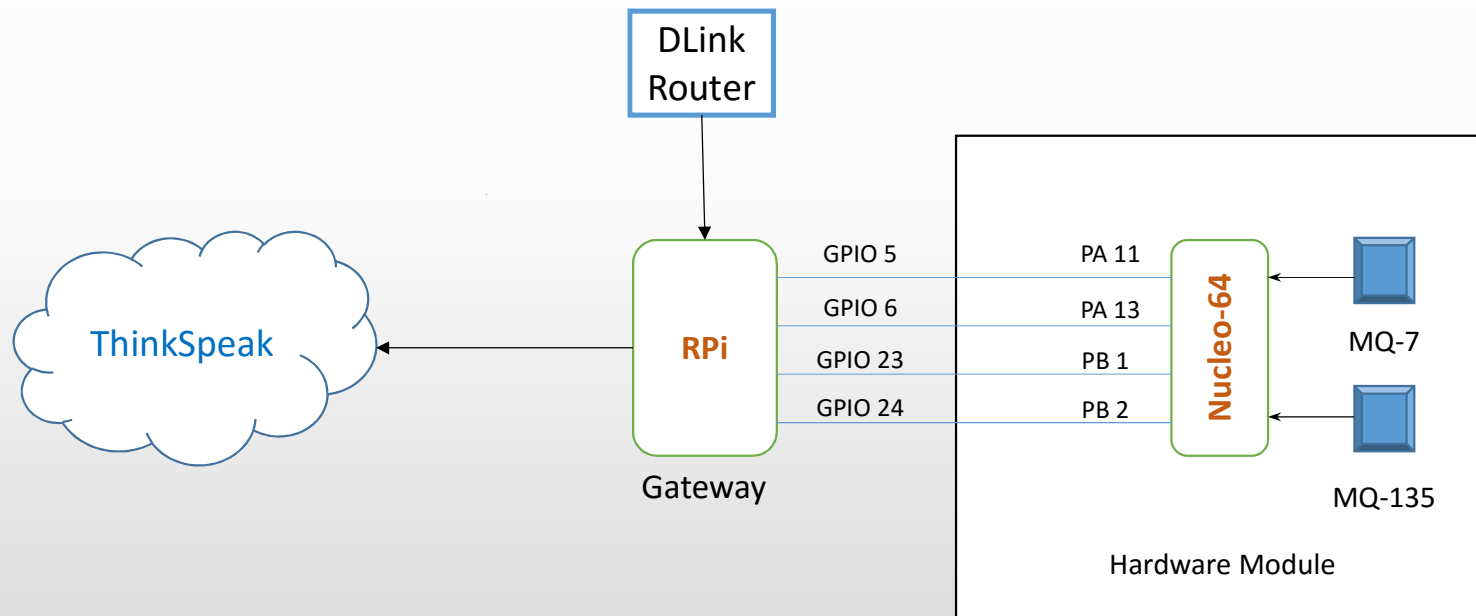
- A Microcontroller Board (STM32 Nucleo-64) with MQ-135, MQ-7 sensors to collect, process and send the data to the central unit.
- The Central unit (Raspberry Pi 3) will work as a Gateway to send the data to the Cloud Platform (ThingSpeak).
- Cloud Platform will perform the real time monitoring of air quality.

PROJECT IMPACT

- This project can be helpful to detect the surrounding air quality and indicatively inform the concerned authorities in near real time.
- The data is sent to the Cloud Platform for trend analysis of air quality during the whole day to take corrective measure.
- It also has the potential to monitor whether the health guidelines (WHO) are properly addressed or not without any human intervention.

BLOCK DIAGRAM

IoT Based Real Time Air Monitoring System



BRIEF DESCRIPTION OF EACH BLOCK

Hardware Module:

It comprises the Nucleo board, MQ-135 sensor, MQ-7 sensor.

Raspberry Pi:

It works as a Gateway to process the data and send it to the Cloud Platform. Here Dlink router is providing the Internet connection.

ThingSpeak:

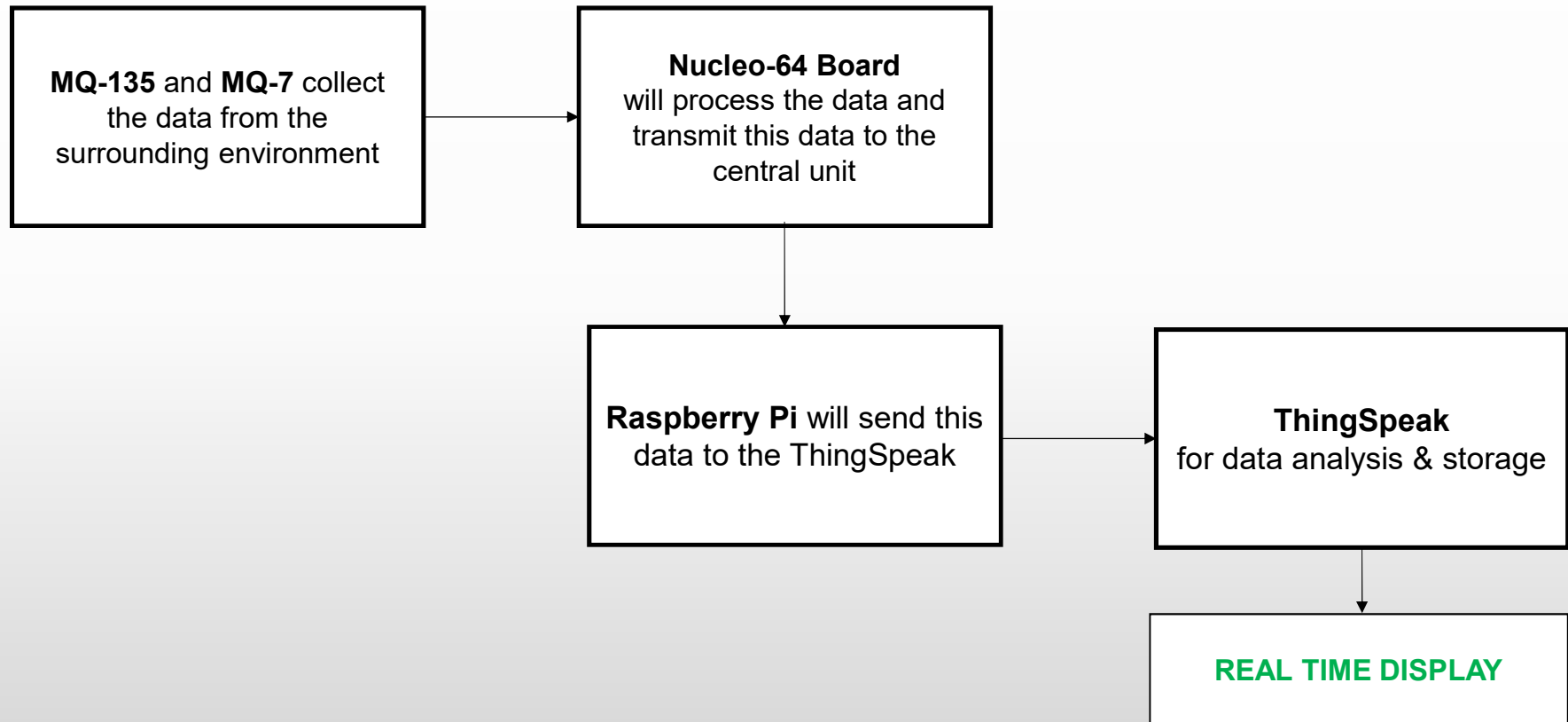
It is a open source cloud platform where we can store our data for trend analysis about the air quality. Through ThingSpeak we can access/download the data for monitoring/analysis purpose.

How this Prototype works:

MQ-135, MQ-7 sensors collect the relative parameters from the surrounding air and capture the data in the Nucleo board.

This data is then send to the cloud platform (ThingSpeak) for analysis and storage via gateway (Raspberry Pi).

FLOW CHART OF AIR MONITORING SYSTEM



TOOLS (H/W & S/W) USED

Hardware:

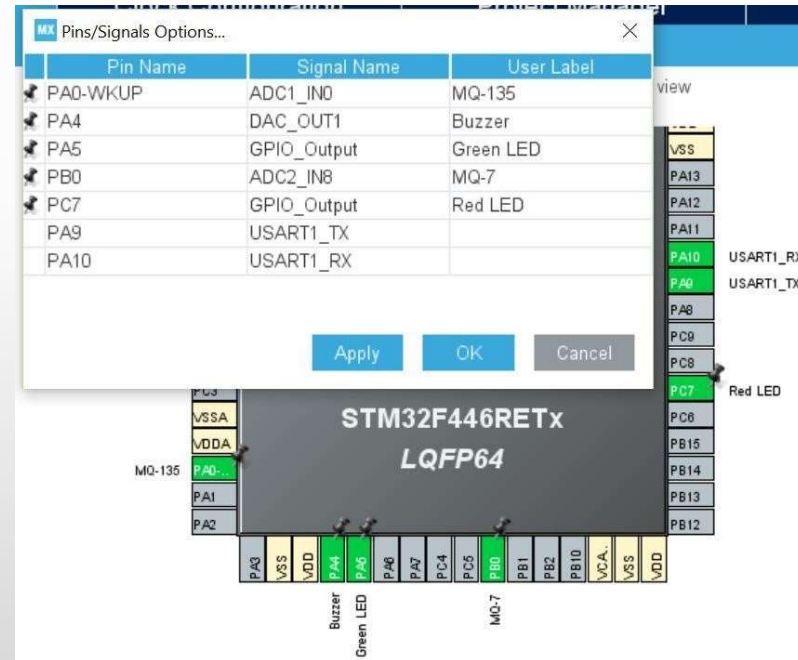
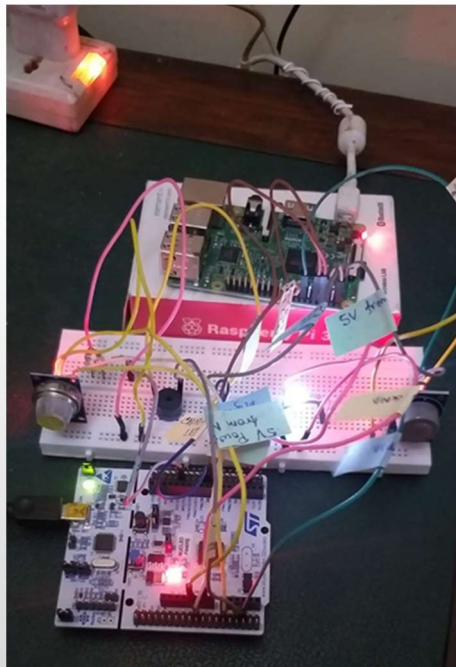
1. STM32F446RET6 Nucleo Board
2. Raspberry Pi 3 Board
3. MQ-135 sensor
4. MQ-7 sensor
5. Jumper Wires
6. D-Link Router

Software:

1. STM32CubeMX
2. Keil IDE
3. VNC Viewer
4. ThingSpeak

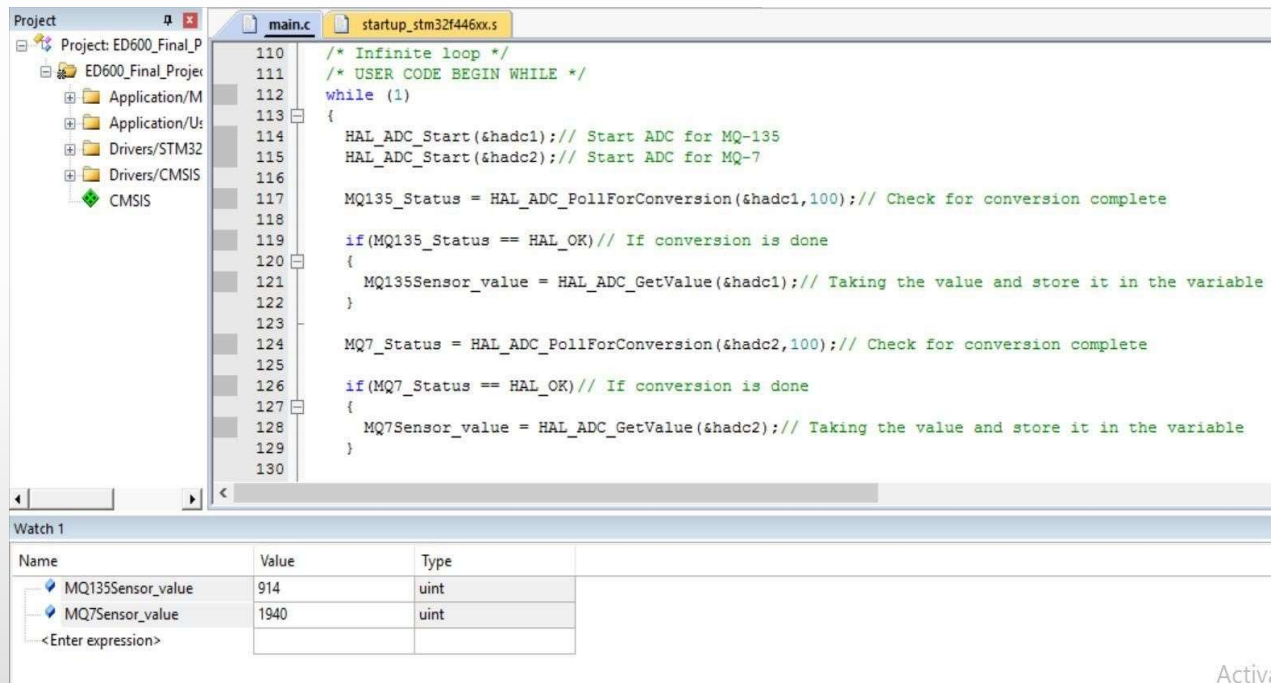
EXPERIMENTAL SETUP

Here I have supplied 5V to MQ-135 sensor from Nucleo-F446RE and MQ-7 sensor to Raspberry Pi. For establishing serial communication between Raspberry Pi and Nucleo board I have connected Pin 8 – GPIO 14 (Tx) of RPi with CN-10 Pin 33 PA-10 (Rx) of Nucleo and Pin 10 – GPIO 15 (Rx) of RPi with CN-10 Pin 21 PA-9 (Tx) of Nucleo board. All other connections are as per Figure 6 pin configuration diagram. At the end ground pin of all the components including boards have been grounded globally.



RESULT

At the time of preparing this report I have received the following outputs:



The screenshot displays the Keil IDE interface. The main editor window shows a C program in `main.c` for the target `startup_stm32f446xx.s`. The code implements an infinite loop that polls two ADCs (MQ-135 and MQ-7) and stores their values in `MQ135Sensor_value` and `MQ7Sensor_value` respectively. The Watch window at the bottom shows the current values of these variables.

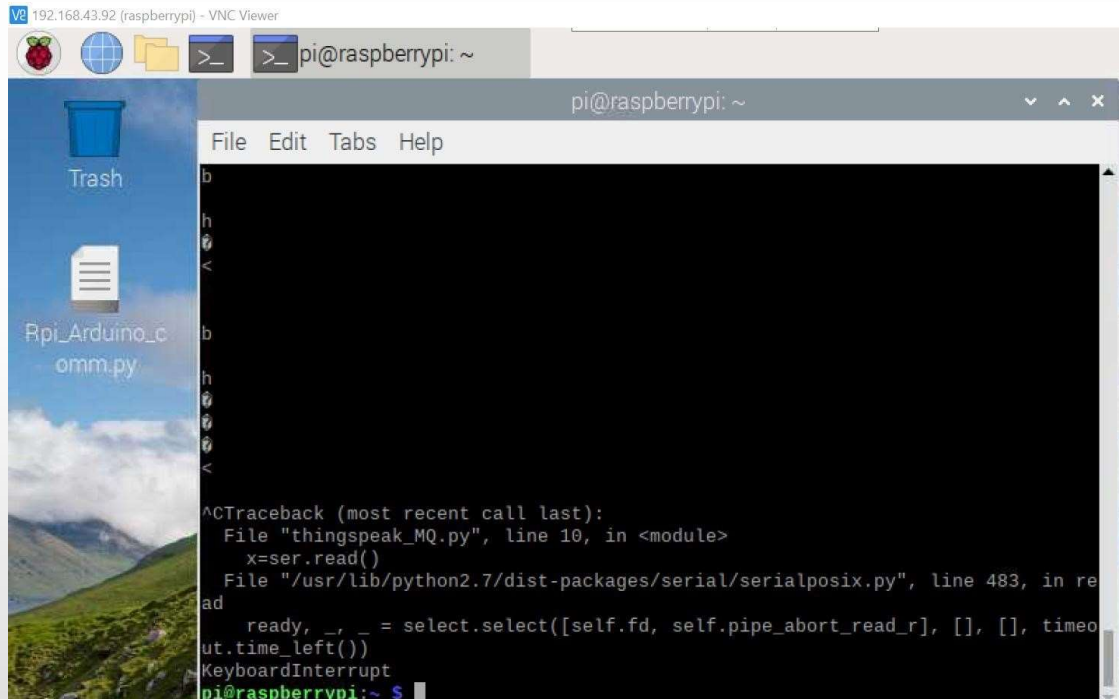
```
110  /* Infinite loop */
111  /* USER CODE BEGIN WHILE */
112  while (1)
113  {
114      HAL_ADC_Start(&hadc1); // Start ADC for MQ-135
115      HAL_ADC_Start(&hadc2); // Start ADC for MQ-7
116
117      MQ135_Status = HAL_ADC_PollForConversion(&hadc1,100); // Check for conversion complete
118
119      if(MQ135_Status == HAL_OK) // If conversion is done
120      {
121          MQ135Sensor_value = HAL_ADC_GetValue(&hadc1); // Taking the value and store it in the variable
122      }
123
124      MQ7_Status = HAL_ADC_PollForConversion(&hadc2,100); // Check for conversion complete
125
126      if(MQ7_Status == HAL_OK) // If conversion is done
127      {
128          MQ7Sensor_value = HAL_ADC_GetValue(&hadc2); // Taking the value and store it in the variable
129      }
130  }
```

Name	Value	Type
MQ135Sensor_value	914	uint
MQ7Sensor_value	1940	uint
<Enter expression>		

Sensor values on watch1 in the Keil

RESULT

At the time of preparing this report I have received the following outputs:



The screenshot shows a VNC Viewer window titled "192.168.43.92 (raspberrypi) - VNC Viewer". The desktop background is a landscape image. On the desktop, there is a "Trash" icon and a file named "Rpi_Arduino_comm.py". A terminal window is open, displaying the following output:

```
pi@raspberrypi: ~  
File Edit Tabs Help  
b  
h  
0  
<  
b  
h  
0  
0  
0  
<  
^CTraceback (most recent call last):  
  File "thingspeak_MQ.py", line 10, in <module>  
    x=ser.read()  
  File "/usr/lib/python2.7/dist-packages/serial/serialposix.py", line 483, in re  
ad  
    ready, _, _ = select.select([self.fd, self.pipe_abort_read_r], [], [], timeo  
ut.time_left())  
KeyboardInterrupt  
pi@raspberrypi:~$
```

Sensor data received at Raspberry Pi terminal

RESULT

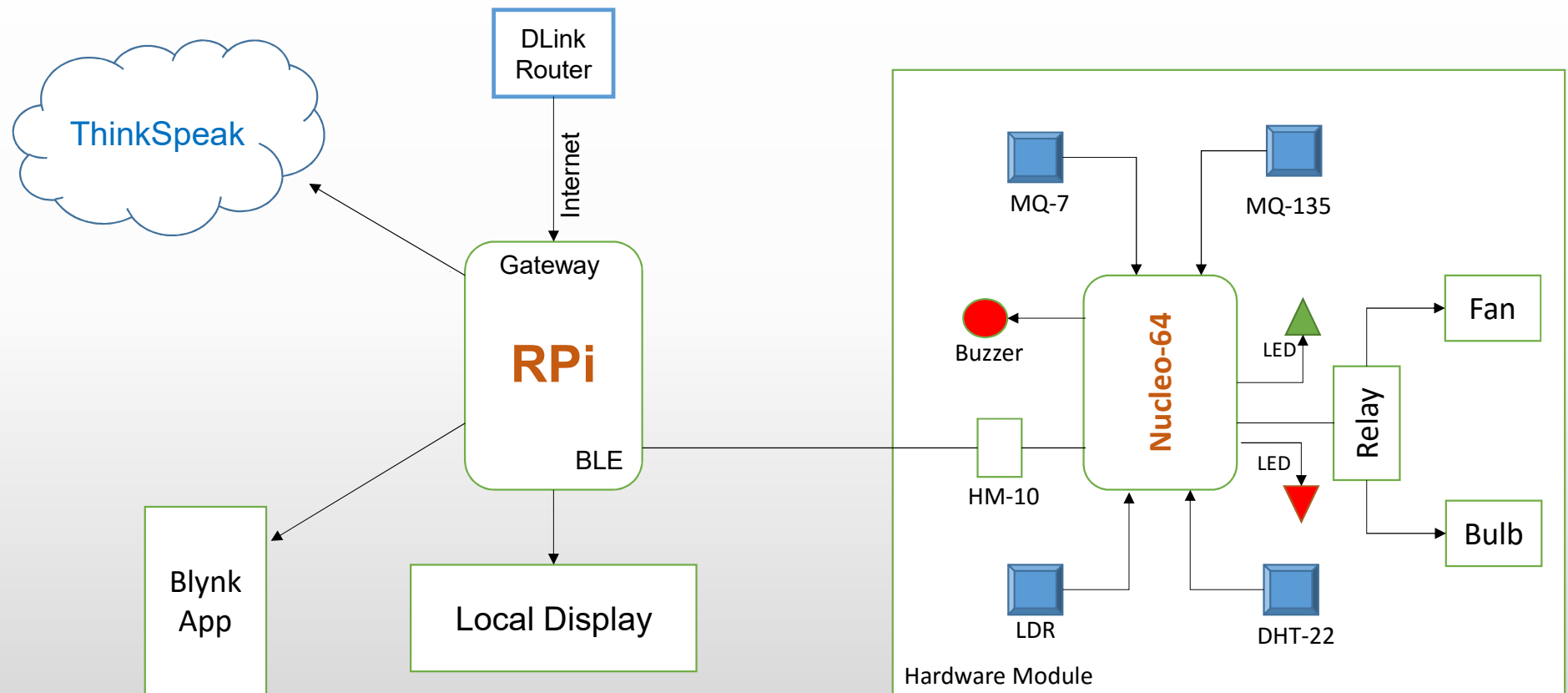
At the time of preparing this report I have received the following outputs:



Data received at ThingSpeak from Raspberry Pi

FUTURE SCOPE OF WORK

Block Diagram of IoT Based Real Time Air Monitoring System



THANK YOU!